SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

Course Code: ITA0512 Course Title: Computer Vision

CO Assessed: CO2 – Imitation (BL3) Assessment: Assessment Tool 1 – Analytical Problem Solving

Weightage: 10% of CIA Total Marks: 100

CO2 Statement:

Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Student Name: RAJASHREE S Reg. No.: 192521079

Section / Batch: Date: 09-08-2026

Code of Conduct

I RAJASHREE S (192521079) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Rajashree S

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

SIMATS ENGINEERING

ASSESSMENT TOOL: Analytical Problem Solving

COURSE NAME: COMPUTER VISION COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Questions:-

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

10	20	30
20	30	40
30	40	50

Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

0	-1	0
-1	4	-1
0	-1	0

to the matrix:

10	10	10
10	20	10
10	10	10

Compute the output at the center.

6. Frequency Domain Filtering

(a) Differentiate low-pass and high-pass filtering in frequency domain.

(b) Given frequency components: F = [100, 80, 40, 20]

Apply a low-pass filter that retains only the first two components. Write the filtered output.

ANSWERS

QUESTION 4:-

(a) Spatial filtering for smoothing is a technique in which each pixel's intensity is replaced with a weighted average of the intensities of its neighboring pixels. Its purpose is to reduce noise and suppress small, unwanted intensity variations in an image, since random noise typically appears as abrupt, high-frequency changes that get blended out by averaging. Smoothing is commonly applied as a pre-processing step before edge detection or segmentation, and it helps produce a cleaner image, though it also blurs fine detail if applied too strongly.

(b) Given the 3×3 region:

10	20	30
20	30	40
30	40	50

An averaging filter replaces the center pixel with the mean of all 9 neighborhood values:

New center value = (10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50) / 9 = 270 / 9 = 30

Result: The new center pixel value is 30. Since the original center pixel (30) already equals the neighborhood average, the smoothing filter leaves the center value unchanged here, reflecting the symmetric, evenly graded nature of the region.

QUESTION 5:-

(a) Sharpening enhances image details by emphasizing the high-frequency components of an image, i.e., the regions where intensity changes rapidly, such as edges and fine textures. Unlike smoothing, which averages neighboring pixels, sharpening filters (such as the Laplacian) compute differences between a pixel and its neighbors to highlight these transitions. The resulting high-frequency response is typically combined with the original image to increase contrast at boundaries, making edges and structures appear crisper. This is useful before feature extraction or object recognition, although it can also amplify noise since noise is itself high-frequency.

(b) Laplacian mask:

0	-1	0
-1	4	-1
0	-1	0

Applied to the region:

10	10	10
10	20	10
10	10	10

Convolving the mask centered on the middle pixel (20):

Output = 4×(20) − 1×(10) − 1×(10) − 1×(10) − 1×(10) = 80 − 40 = 40

Result: The output at the center pixel is 40. This positive Laplacian response shows that the center pixel (20) is brighter than its four immediate neighbors (10 each), i.e., it lies at a local intensity peak — exactly the type of feature the Laplacian mask is designed to detect and enhance.

QUESTION 6:-

(a) In the frequency domain, an image's Fourier components represent how intensity varies at different spatial frequencies. Low frequencies correspond to smooth, slowly varying regions (overall shape and background), while high frequencies correspond to sharp transitions such as edges, fine textures, and noise. A Low-Pass Filter (LPF) retains the low-frequency components and suppresses the high-frequency ones, which smooths the image and reduces noise but also blurs edges and fine detail. A High-Pass Filter (HPF) does the opposite — it retains high-frequency components and suppresses low-frequency ones, sharpening the image and emphasizing edges, though it also amplifies noise and removes smooth shading information.

(b) Given Frequency Components: F = [100, 80, 40, 20]

Assume the LPF retains only the first two (lowest-frequency) components and removes the remaining higher-frequency components 40 and 20.

Filtered Output: F' = [100, 80, 0, 0]

The high-frequency components responsible for fine detail and noise are removed, while the low-frequency components, which represent the overall smooth structure of the signal/image, are retained. As a result, the output represents a smoothed version with reduced high-frequency variation.